A pause, not an end: mining after Operation Mercurio

What it means for FZS's work in Tambopata. Prepared for the Frankfurt Zoological Society.

Group Giant Anteater · Ruby Ann Grambauer, Sebastian Schmid, Laurens van der Lely, Omkar Mate, Dhruv Chaudhari

Feb 2019	30 m	6
Operation Mercurio starts	satellite pixels since 1985	ways we tested for cause



Photo: La Pampa, Madre de Dios, 2019 (army operation). Eliftoledo, Wikimedia Commons, CC BY-SA 4.0, cropped.

Anteaters eat 35,000 ants a day, and never destroy a nest

60 cm

OF TONGUE

Flicks up to 150 times a minute.

Smithsonian's National Zoo; National Geographic

1 minute

PER MOUND

Then it moves on, and comes back later.

National Geographic



Giant anteater at a termite mound

Mining in La Pampa did the same: it moved on, then came back.

FZS pays for operations. What happens when one ends?

Net mining additions, ha/yr · Tambopata buffer zone, La Pampa, Madre de Dios · 2016–18 vs 2019–21 vs 2022–25

WHAT WE FOUND

1 · IT FELL, LOCALLY

1,640 → 163

ha/yr, Tambopata buffer zone. 25%–47% of the fall: pits going out of use

2 · IT MOVED

60 → 395

ha/yr just across the road, in the legal corridor

3 · IT CAME BACK

1,829

ha/yr in the later years (2022–25), much of it on old pits

All three: description. Net additions = new land + re-mined − stopped, from yearly 30-m satellite maps.

DESCRIPTION

La Pampa sits one road away from legal mining land

Madre de Dios, Peru · step 1 of 4 · zones as used on every slide

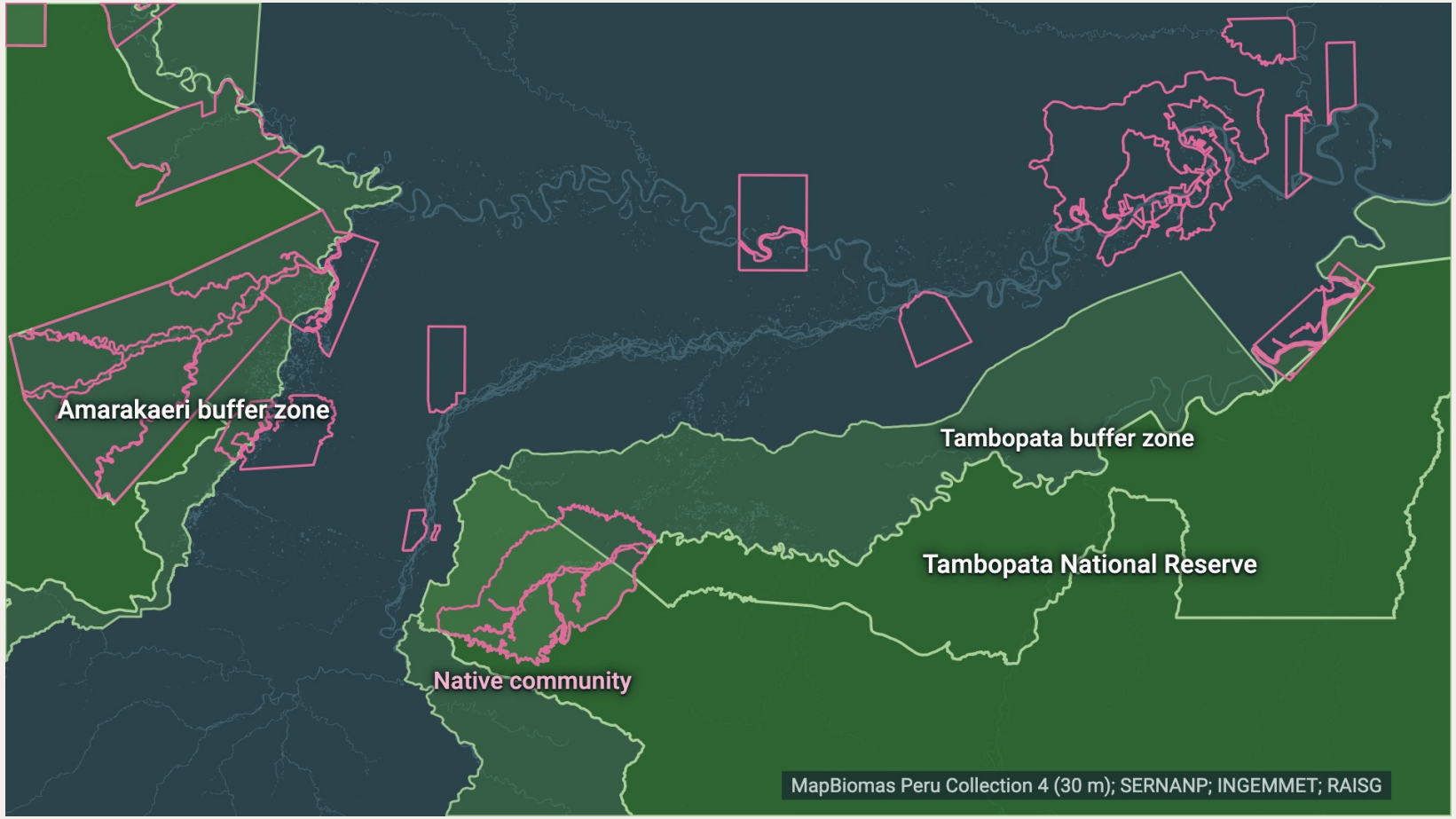
1 **Protected areas**

Tambopata reserve, its buffer zone and the Amarakaeri buffer zone. Pink: communities.

2 The Interoceanic Highway

3 Where mining can be legal

4 The target



DESCRIPTION

La Pampa sits one road away from legal mining land

Madre de Dios, Peru · step 2 of 4 · zones as used on every slide

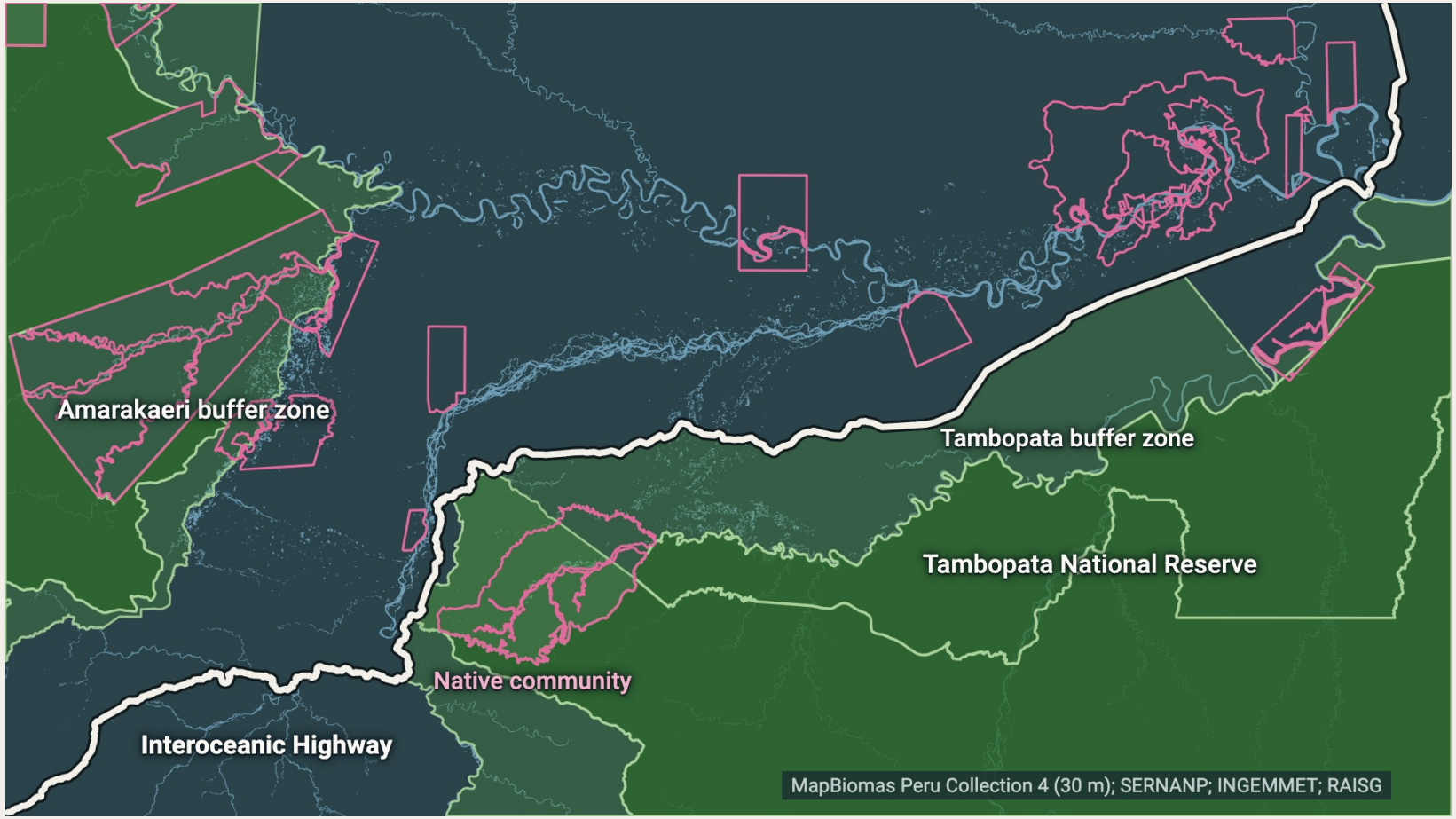
1 Protected areas

2 The Interoceanic Highway

One road runs along the buffer zone and splits it from the land to the north.

3 Where mining can be legal

4 The target



DESCRIPTION

La Pampa sits one road away from legal mining land

Madre de Dios, Peru · step 3 of 4 · zones as used on every slide

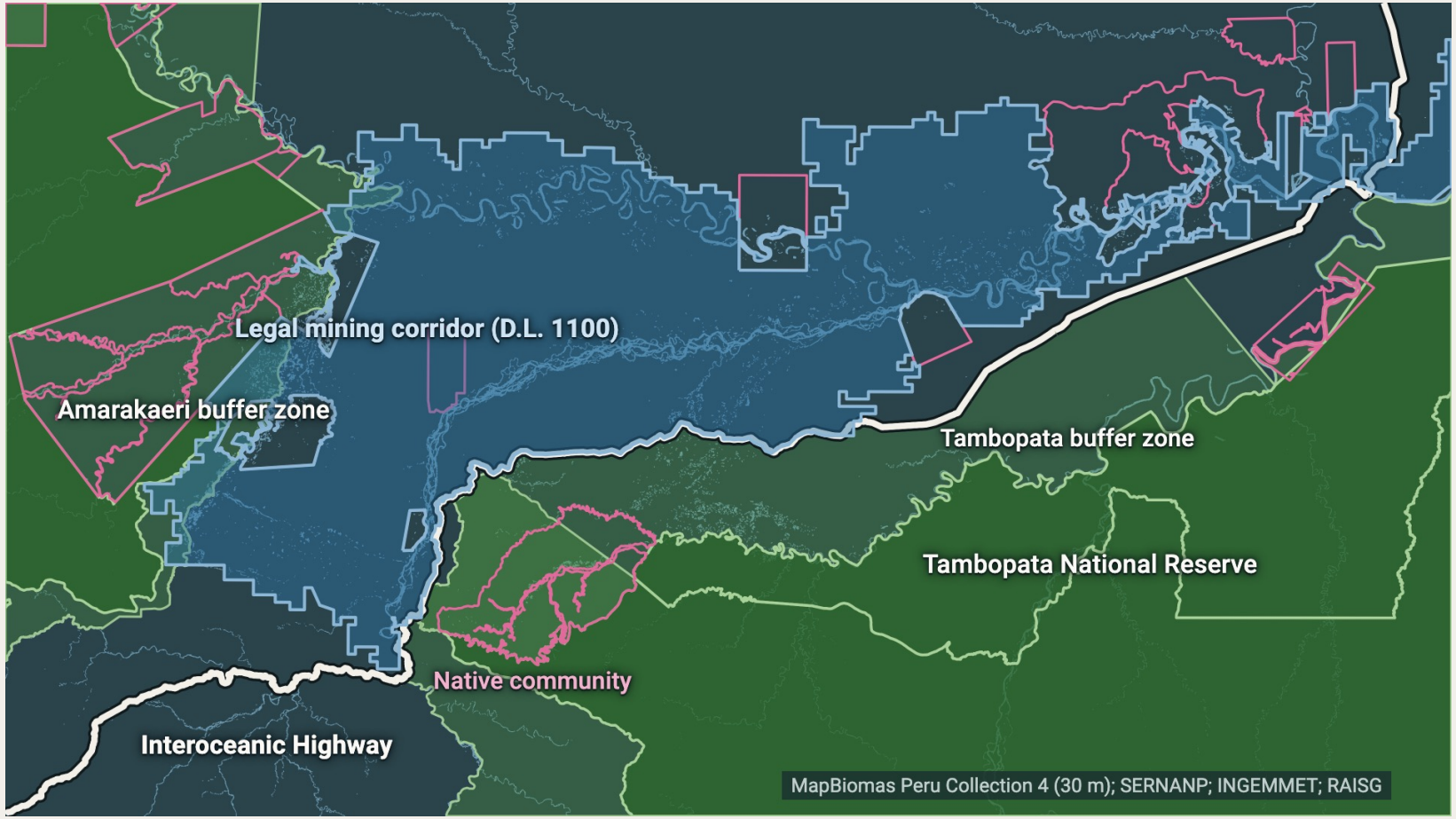
1 Protected areas

2 The Interoceanic Highway

3 Where mining can be legal

The legal mining corridor, north of the road. South of it, in the buffer zone: a crime.

4 The target



DESCRIPTION

La Pampa sits one road away from legal mining land

Madre de Dios, Peru · step 4 of 4 · zones as used on every slide

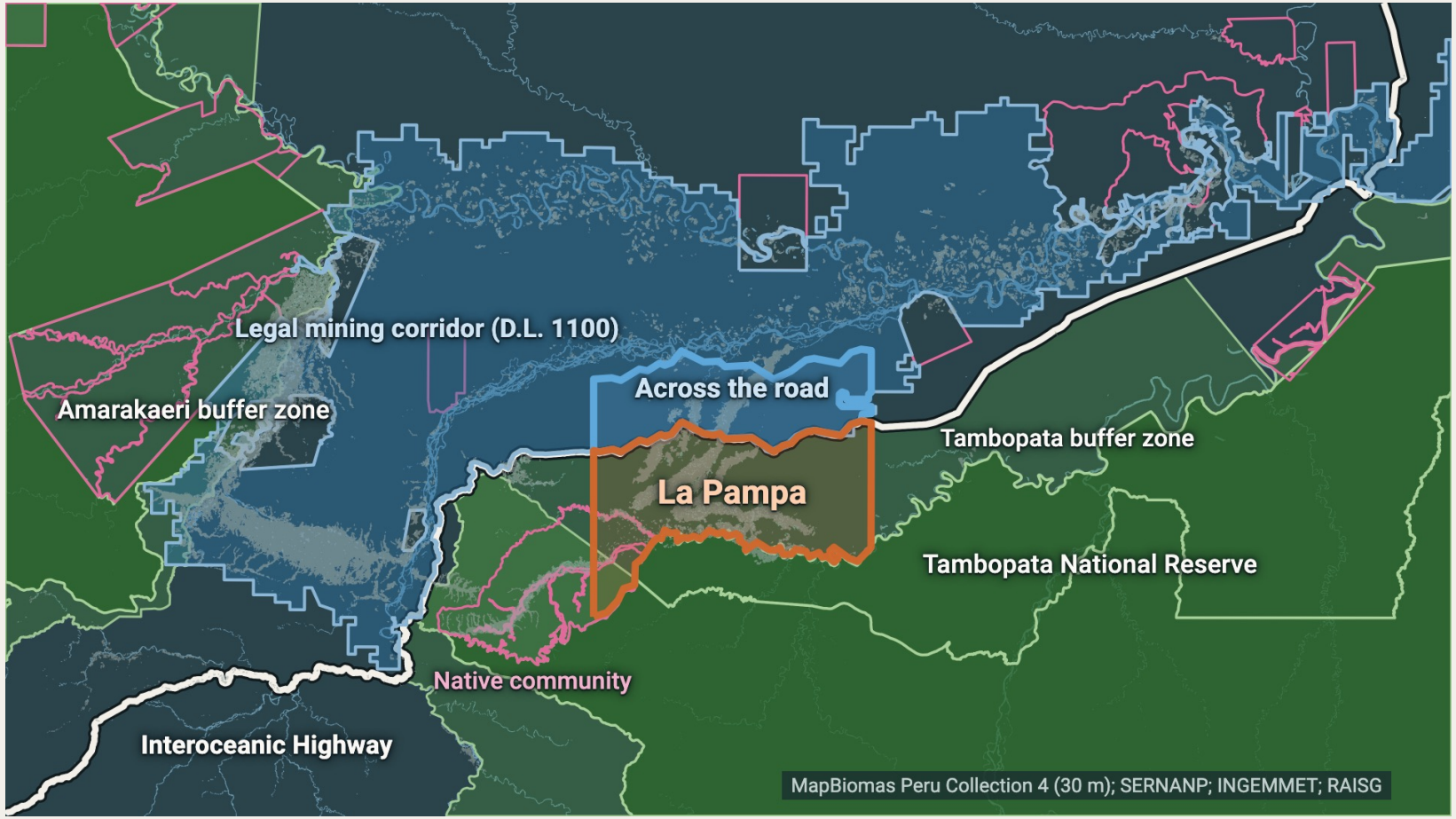
1 Protected areas

2 The Interoceanic Highway

3 Where mining can be legal

4 The target

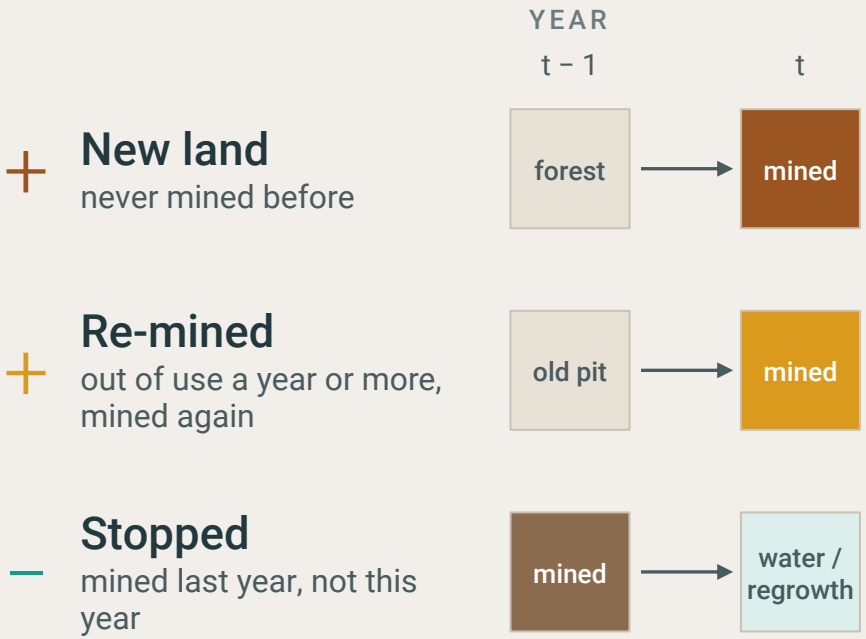
La Pampa, Mercurio's target from Feb 2019, and the corridor strip across the road.

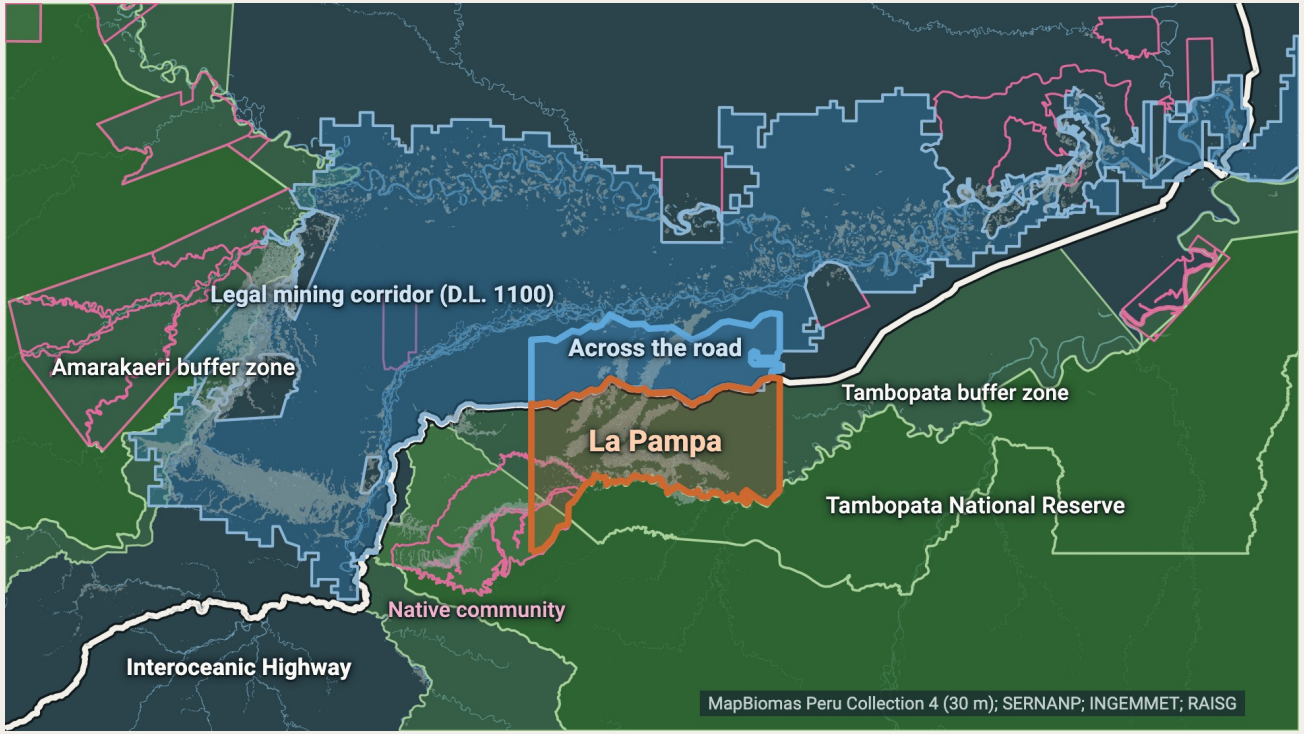


DESCRIPTION

One measure on every slide: net mining additions

Net additions (ha/yr) = new land + re-mined − stopped · per 30-m pixel, year t vs the years before · MapBiomas Peru Collection 4, 1985–2025





Same zones everywhere: Tambopata buffer zone (whole, incl. Kotsimba) · La Pampa sector (its 70.15–69.80° W part) · corridor across the road (D.L. 1100, ≤ 10 km north of the highway) · Amarakaeri buffer zone (less treated) · Madre de Dios.

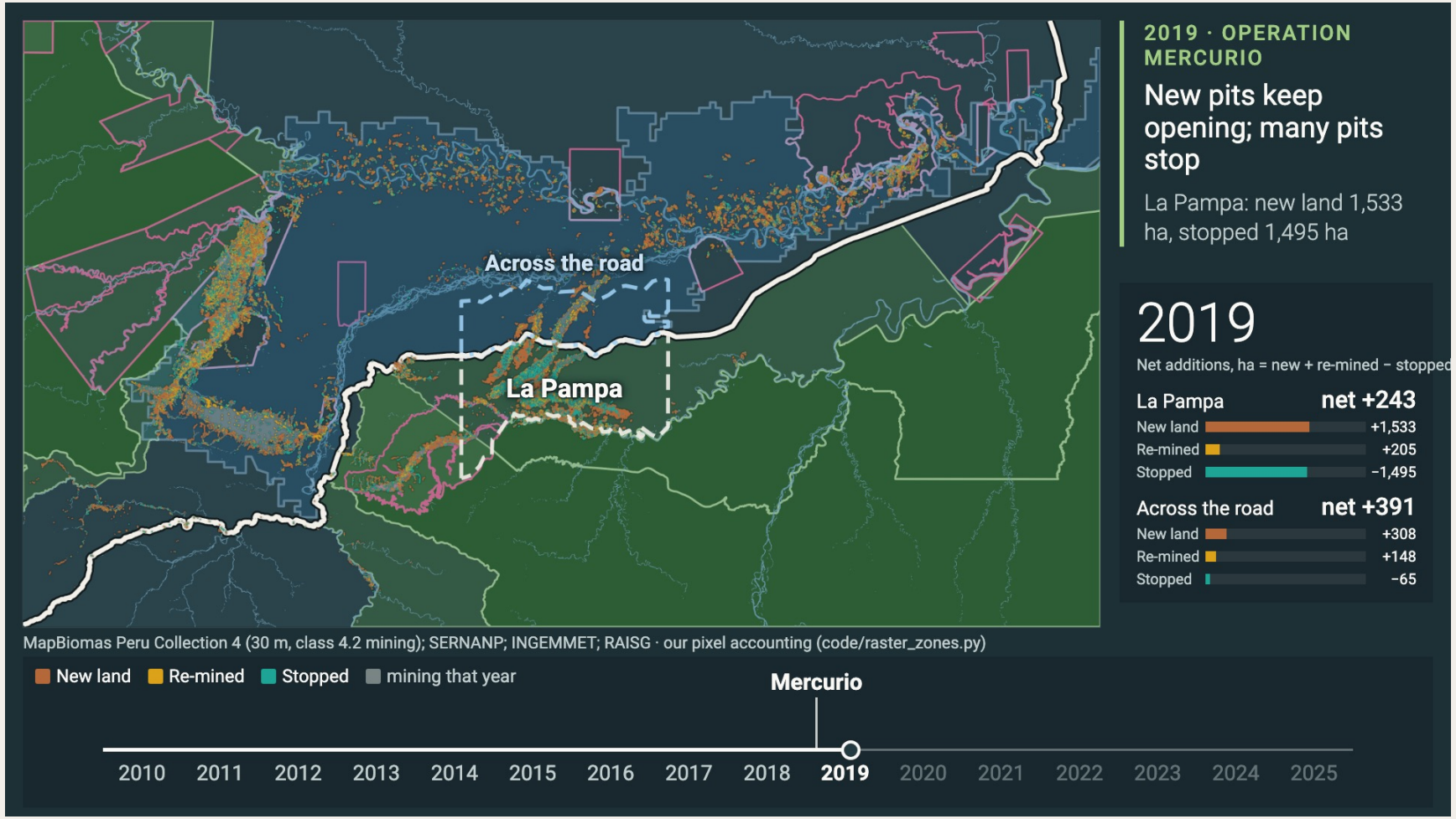
Net = new + re-mined − stopped

= mining area this year − last year. Hectares are a proxy for mining activity, not for miners.

DESCRIPTION

2010–2025: new pits, old pits re-mined, pits stopping

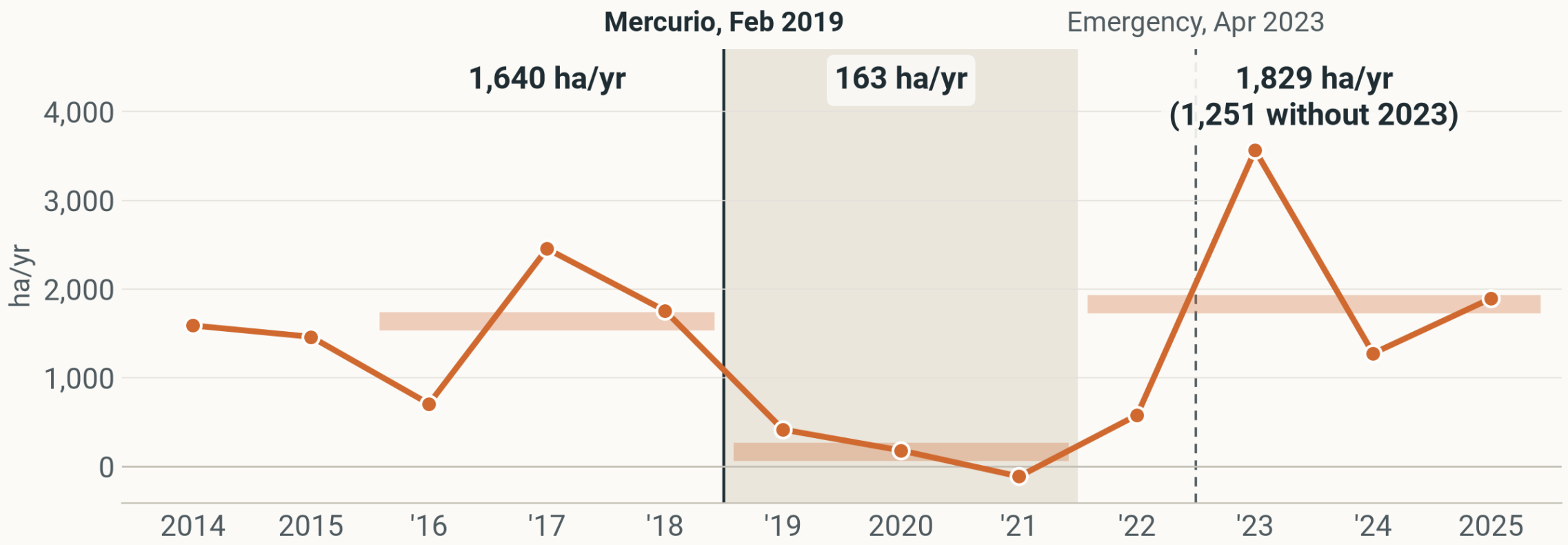
Net mining additions as three flows on the map: new land (brown), re-mined (amber), stopped (teal) · counter: net ha per year, La Pampa sector and corridor across the road



DESCRIPTION

Net mining fell 90%, then came back

Net mining additions, ha/yr · Tambopata buffer zone · 2014–2025 · thick bars and numbers: means for 2016–18, 2019–21 (shaded) and 2022–25

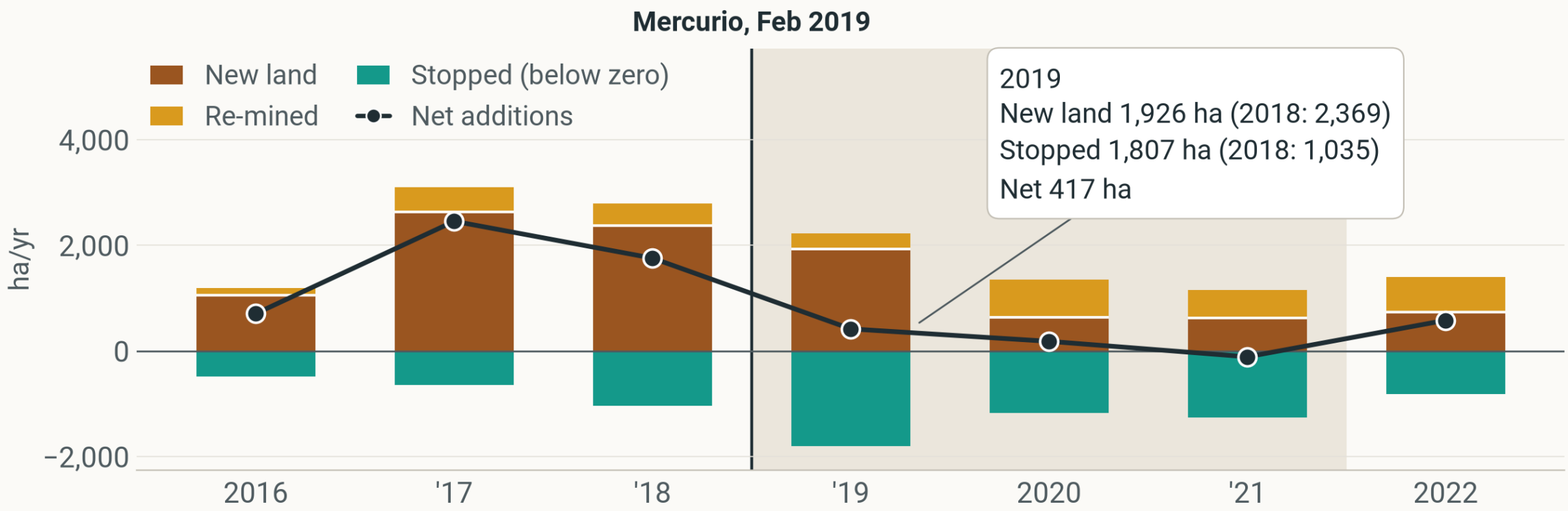


Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. Official SERNANP buffer-zone polygon.
Satellites cannot see river dredges or mercury.

DESCRIPTION

2019: new pits kept opening, old ones went quiet

Net mining additions and parts, ha/yr · Tambopata buffer zone · 2016–2022 · more stopping explains 25%–47% of the 2016–18 → 2019–21 fall; in Amarakaeri stopped land stayed flat (542 → 584 ha/yr)

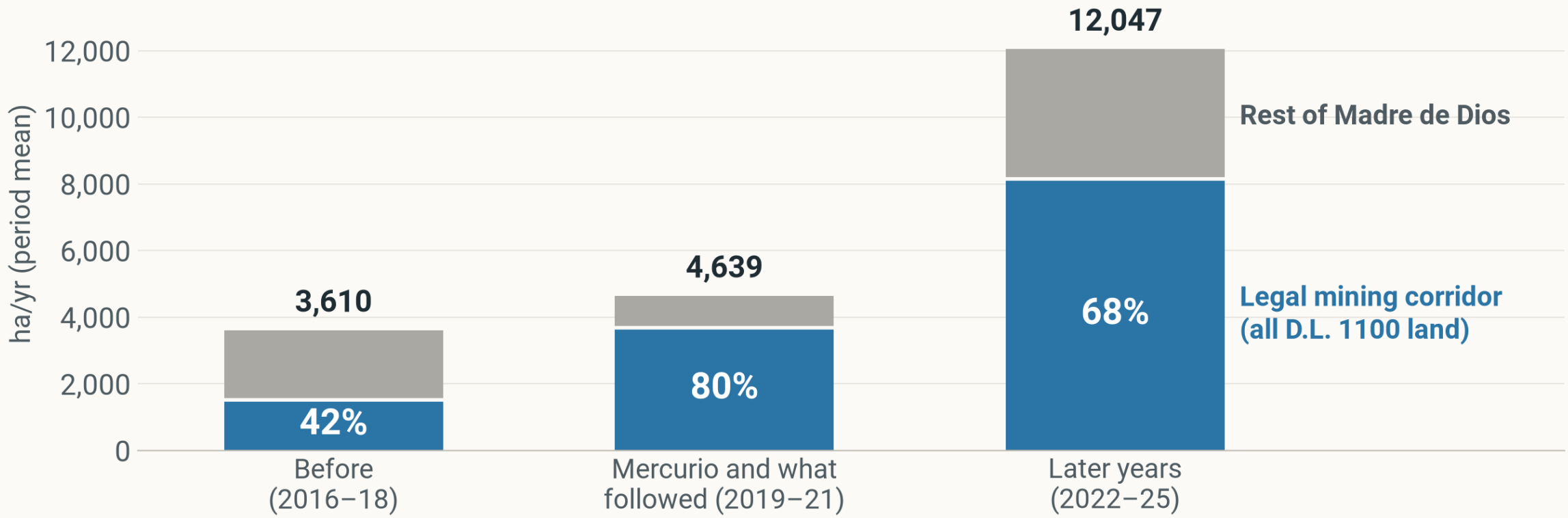


Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. New land = mined for the first time since 1985; re-mined = mined again after a gap; stopped = left the mining class (water, regrowth). Satellites cannot see river dredges or mercury.

DESCRIPTION

The region kept growing, mostly in the legal corridor

Net mining additions, ha/yr, period means · Madre de Dios · % = the legal corridor's share of the department's net additions (period sums) · legal corridor = all D.L. 1100 land, not only the strip across the road

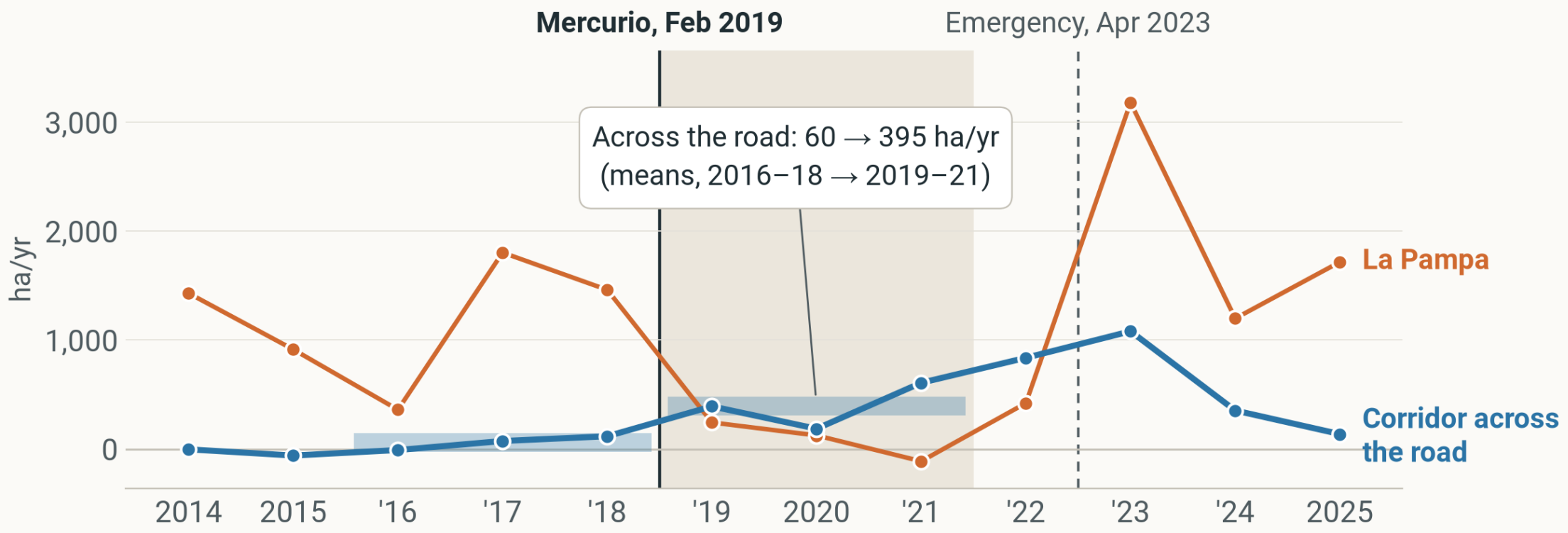


Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. Legal mining corridor: D.L. 1100 polygon (INGEMMET), native communities excluded; legality is by location, the map class cannot tell legal from illegal. Satellites cannot see river dredges or mercury.

DESCRIPTION

Across the road, net mining rose more than six-fold

Net mining additions, ha/yr · corridor across the road (legal strip ≤ 10 km north of the highway) vs La Pampa sector (1,209 → 85 ha/yr) · 2014–2025 · thick bars: means

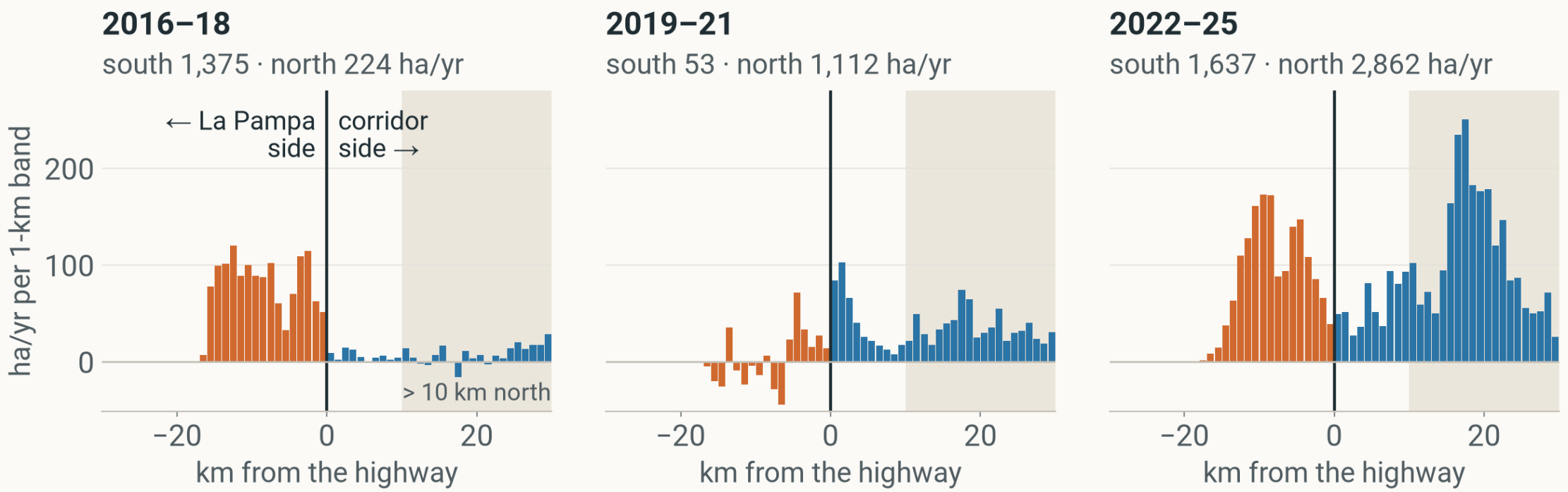


Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. La Pampa sector = Tambopata buffer zone 70.15–69.80° W (our boundary); corridor across the road = D.L. 1100 land at the same longitudes. We cannot show it was the same miners. Satellites cannot see river dredges or mercury.

DESCRIPTION

2019–21: mining fell south of the highway, rose north

Net mining additions per 1-km band, ha/yr · La Pampa's longitudes, 30 km either side of the Interoceanic Highway · means for 2016–18, 2019–21, 2022–25 · shaded: more than 10 km north

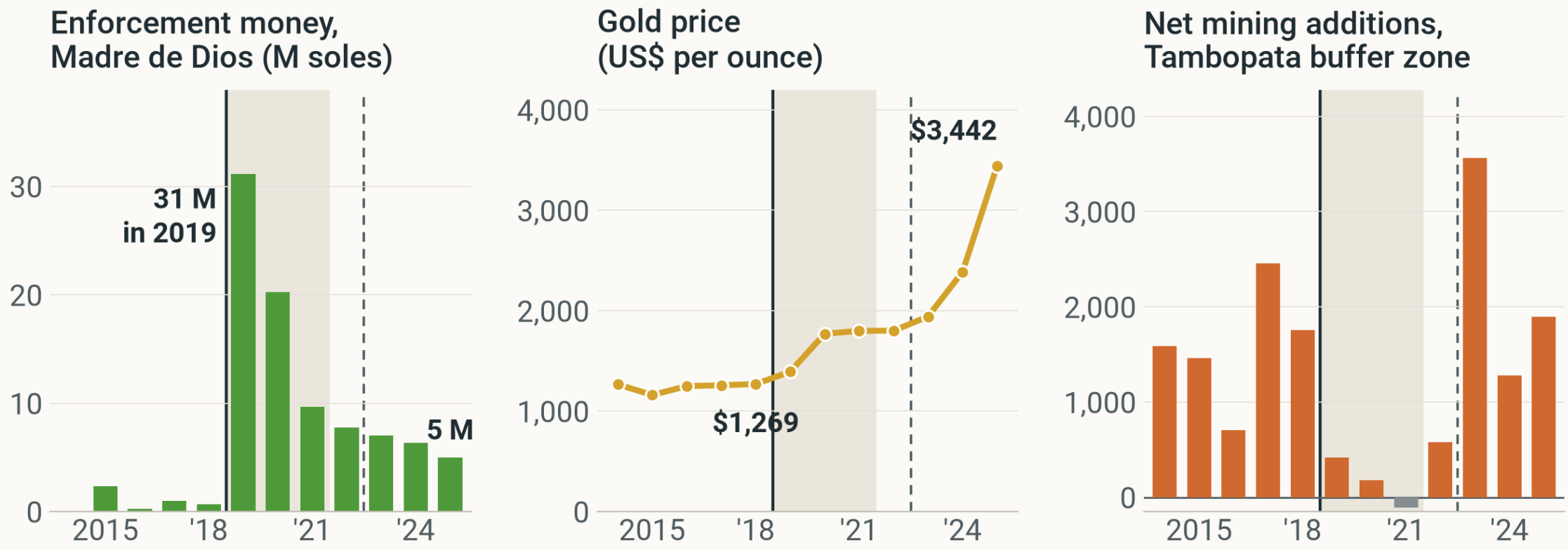


Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. Bands are measured north–south from the highway at 70.15–69.80° W; all land in each band. Description: we cannot show it was the same miners. Satellites cannot see river dredges or mercury.

DESCRIPTION

Money faded after 2019, while gold rose 2.7-fold

Same years in each panel, 2014–2025 · PP 0128 executed spending booked in Madre de Dios (a lower bound) · gold price, nominal · net mining additions, Tambopata buffer zone · shaded: 2019–21 · context, not a test

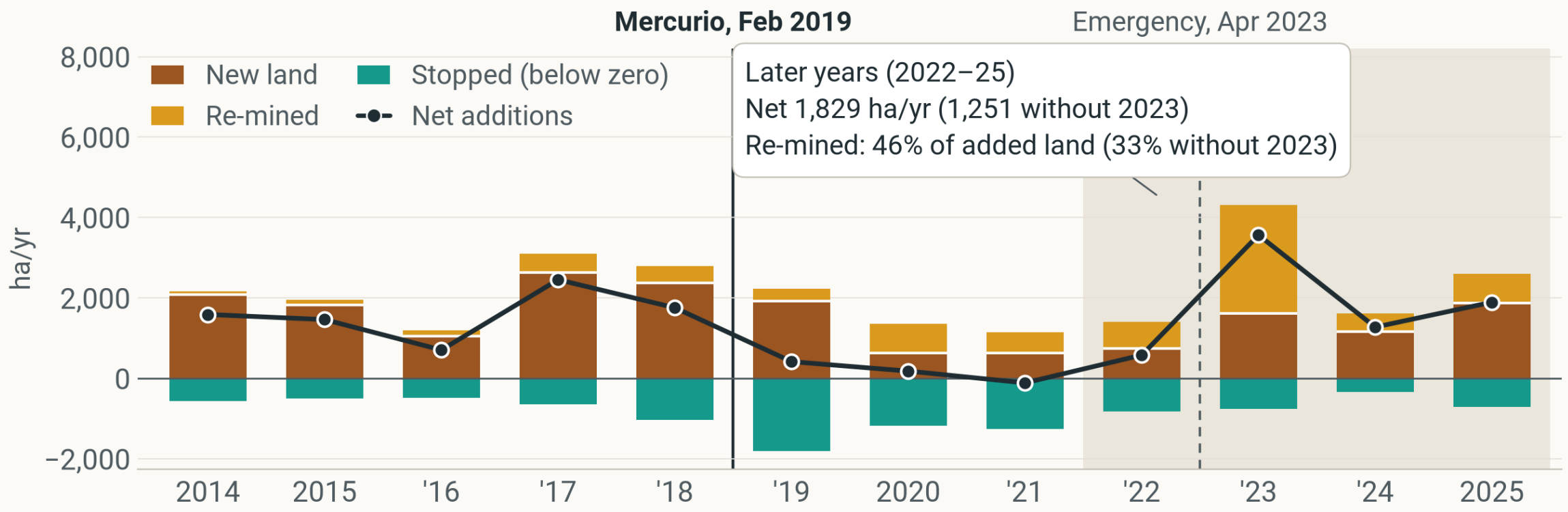


Source: MEF Consulta Amigable, PP 0128 devengado, Madre de Dios; World Bank Pink Sheet (nominal gold price, annual mean); MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. No correlation shown: 11 years, and money follows mining.

DESCRIPTION

It came back, partly on old pits (33–46%)

Net mining additions and parts, ha/yr · Tambopata buffer zone · 2014–2025 · shaded: later years (2022–25) · share = re-mined ÷ (new + re-mined)

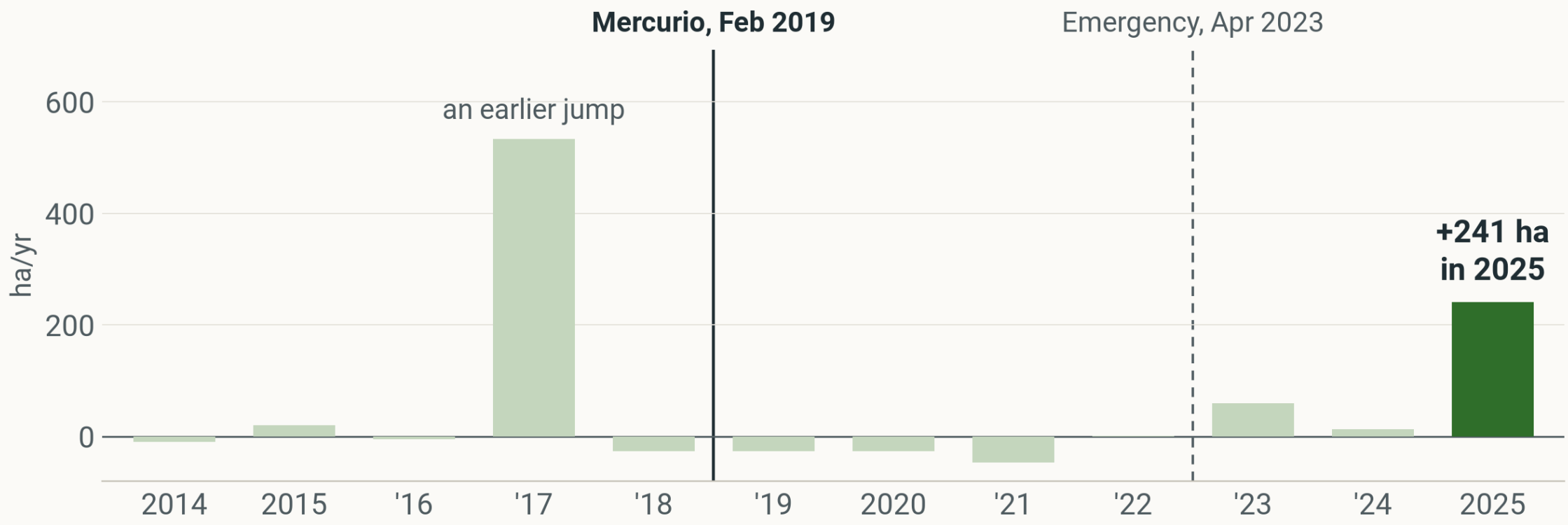


Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. 2023 is a spike in MapBiomas only (forest-loss alerts and Amazon Mining Watch are flat), so later years are shown with and without it. MAAP #193 (2023) reports miners back on degraded land. Satellites cannot see river dredges or mercury.

DESCRIPTION

In 2025, mining inside the reserve jumped 241 ha

Net mining additions, ha/yr · Tambopata National Reserve (inside the reserve, not the buffer zone) · 2014–2025



Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. Platform check: 777 ha of mining inside the reserve in 2025 (protected-area statistics). MAAP #241 (Pacsi et al., 2026) reports ~500 ha inside the reserve, late 2025 to Feb 2026. Satellites cannot see river dredges or mercury.

CAUSAL · CONDITIONAL

Six methods: two hold up, none is clean

Each builds a stand-in for the Tambopata buffer zone without Mercurio · all six point down · green = holds up, in net additions, in hectares only

Compare with other zones

Tambopata vs 14 other Peruvian buffer zones, before and after

HOLDS UP · IN HECTARES ONLY

A weighted copy

a mix of zones built to track Tambopata before 2019

NO GOOD COPY BEFORE 2019

Twins in Brazil

look-alike mining areas picked by fixed rules

DIFFERENT TRENDS BEFORE 2019

Either side of the road

land just north vs just south of the highway

TOO NOISY TO TELL

Matching, 5 ways

look-alike areas picked by algorithm

CAN'T BEAT CHANCE

Weighted zones and years

zones and years weighted to match Tambopata's past

HOLDS UP · IN HECTARES ONLY

CAUSAL · CONDITIONAL

What we can say about cause

If the other zones show what Tambopata would have done, Mercurio and the operations that followed lowered net mining additions by 1,453 to 1,663 ha a year in 2019–21.

Hectares only

Largest fall of 15 zones; for its size, 3rd–6th of 7, depending on the scaling.

A quarter to a half is pits stopping

25%–47% of the fall is more land going out of use, most of it in 2019.

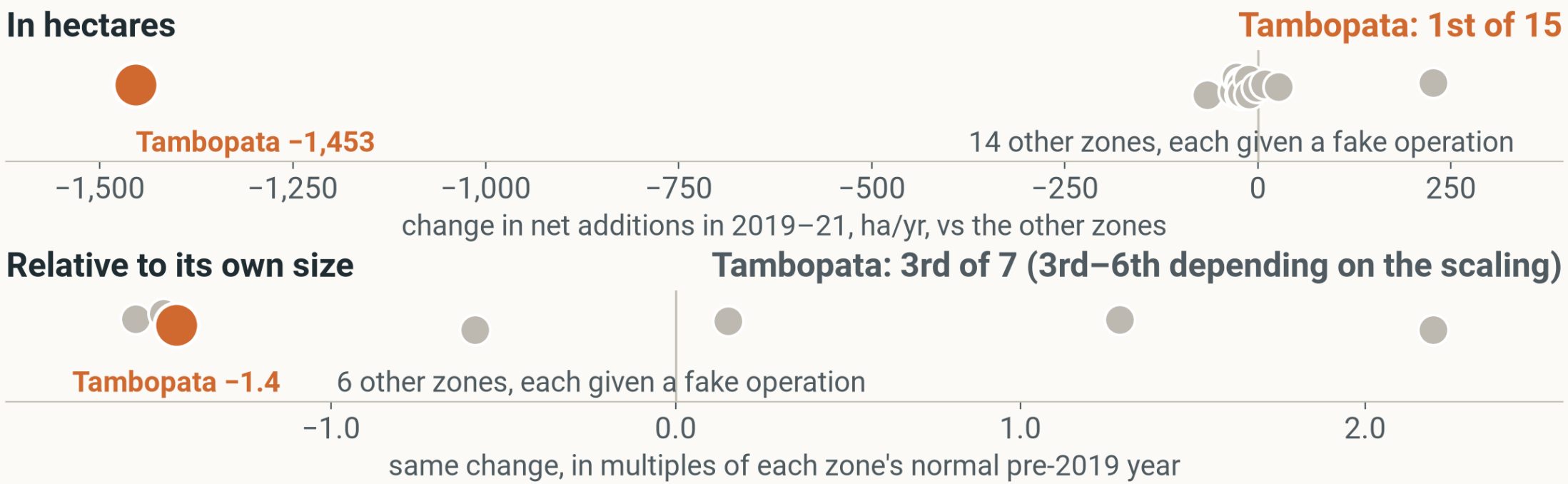
COVID overlaps

New land fell only from 2020, the first COVID year.

CAUSAL · CONDITIONAL

The fall stands out in hectares, not for its size

Each grey dot: one Peruvian buffer zone given a fake operation in 2019 · orange: the Tambopata buffer zone's real 2019–21 change · net mining additions, after each zone's level and each year's shared change



Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. 15 Peruvian buffer zones with mining before 2019 (scaled: 7 with more than 10 ha/yr). Tambopata removed from each fake zone's comparison pool. A rank shows whether the real result stands out; with one target it is not a significance test.

DESCRIPTION

Most biases make the fall look bigger

Net mining additions measure land, not people, rivers or mercury · direction of each bias for the 2019–21 fall, Tambopata buffer zone

SATELLITES CAN'T SEE

River dredges

148 → 598 in La Pampa, 2021–23 (MAAP #193)

Mercury

in rivers and fish

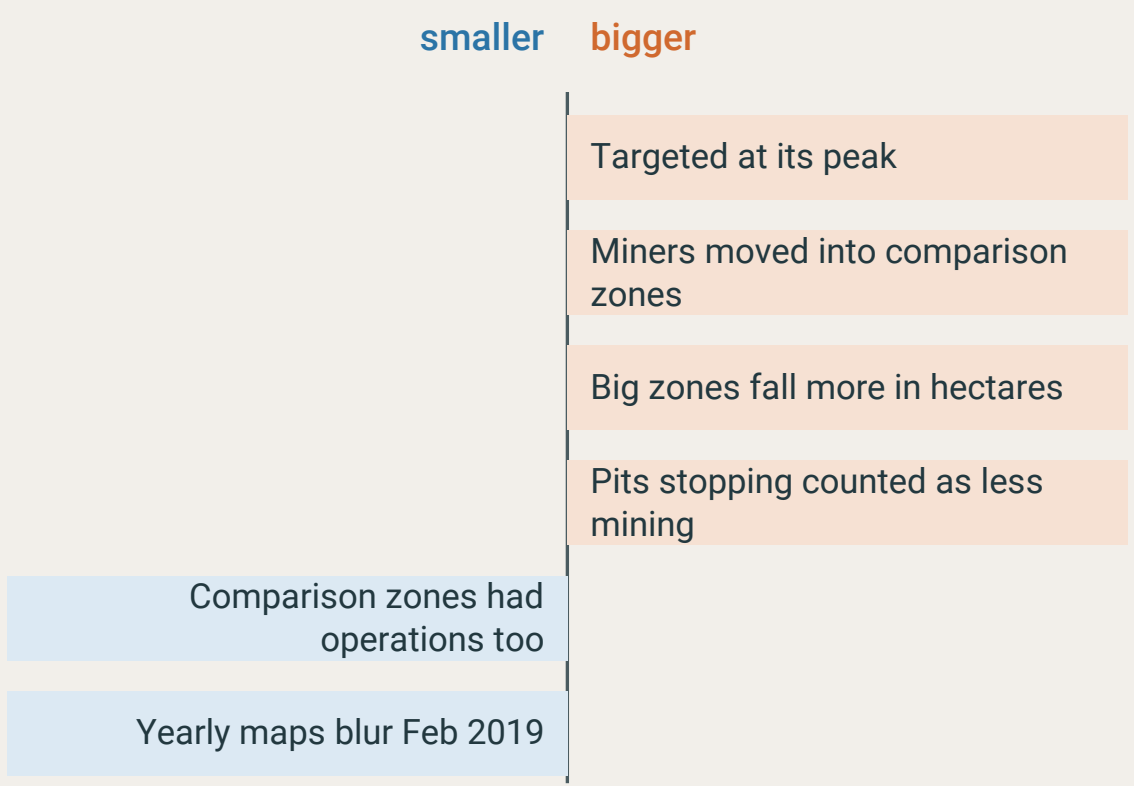
People

hectares, not miners

Map flicker

37% of land that stopped in 2019–21 was mined again within 2 years

EACH BIAS MAKES THE 2019–21 FALL LOOK…



Read the fall as an upper bound, the return as a lower bound.

PREDICTION

Plan for the months after the operation ends

Informed expectations from one case, not proven rules · net mining additions, Tambopata buffer zone, corridor across the road, reserve

Stay after the sweep

163 → 1,829

ha/yr, 2019–21 → later years, Tambopata buffer zone. Fund presence, not one-off sweeps.

Watch the ring around it

60 → 395

ha/yr just across the road. Near-real-time forest-loss alerts on a ring of a few km; add old pits.

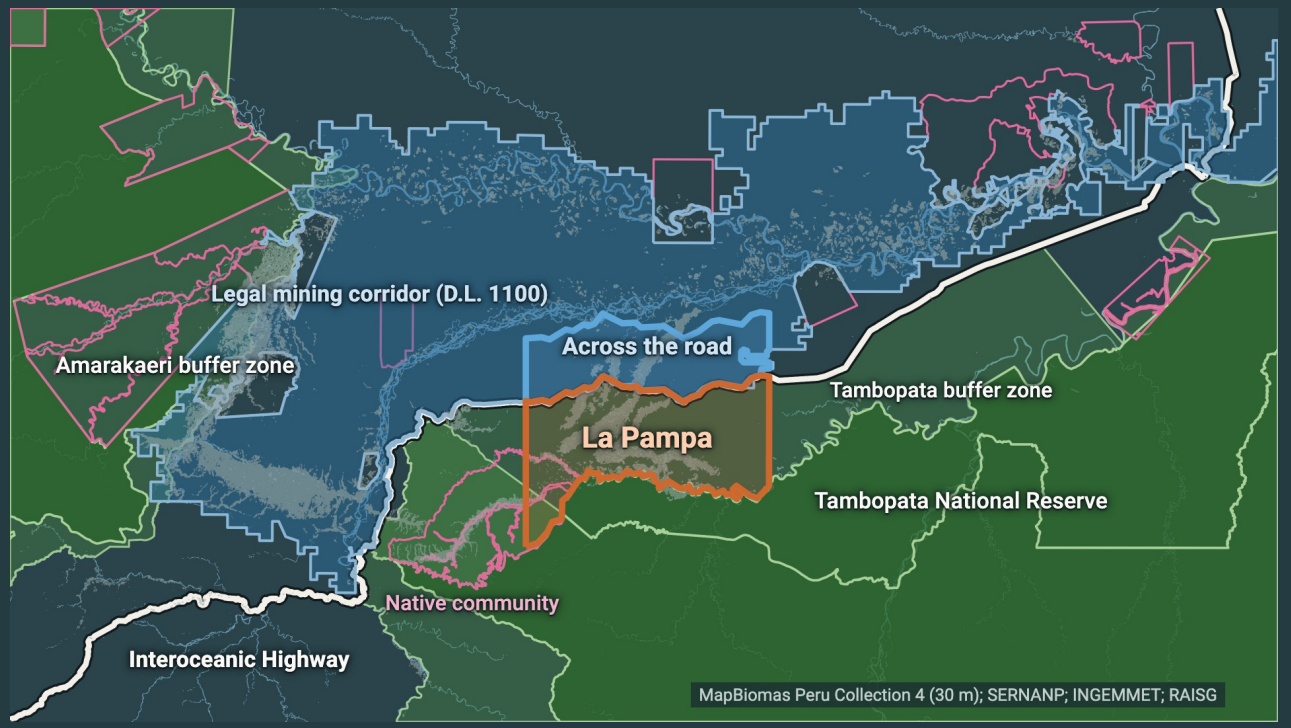
Guard the reserve's edge and river

91%

of first-time mining inside the reserve along its buffer-zone border (2000–25) lies within 1 km of the edge.

Questions?

Like the anteater: a pause, not an end.



Group Giant Anteater · Ruby Ann Grambauer, Sebastian Schmid, Laurens van der Lely, Omkar Mate, Dhruv Chaudhari

For questions

Six methods · one causal range · stopped land · road and cells · legality · declared gold · money · risk map · twins · Brazil · map video · one measure · definitions

Six methods in detail

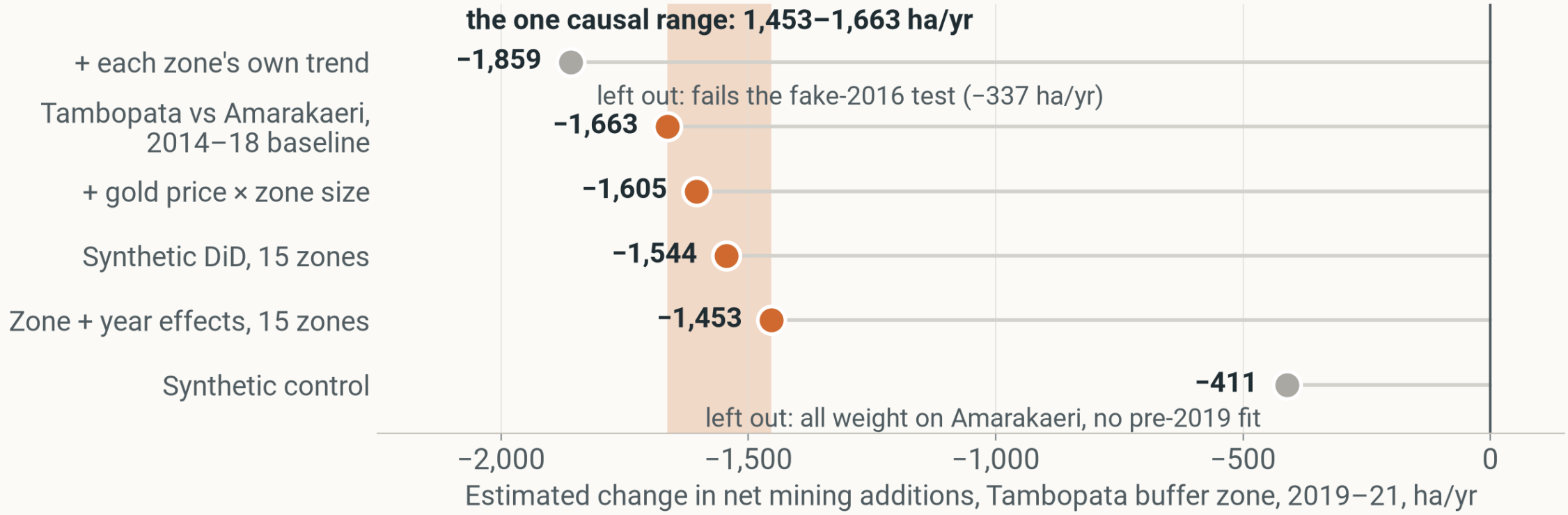
Net mining additions, Tambopata buffer zone, 2019–21 vs before, ha/yr unless stated · all conditional on the comparison

METHOD: WHAT IT DOES	RESULT	VERDICT
1 Difference-in-differences: Tambopata's change minus the comparison zones' change	−1,453 (vs 14 zones) to −1,663 (vs Amarakaeri), 2014–18 base; size rank 3rd–6th of 7	Holds up in ha
2 Synthetic control: a weighted mix of zones that tracks Tambopata before 2019	−411; all weight on Amarakaeri, no pre-2019 fit	No good copy
3 Rule-based twins: Brazilian areas picked by Ruby's four rules	Tambopata at 16%–41% of their pace (Amazon Mining Watch, another product)	Different trends
4 Spatial discontinuity: land just north vs just south of the road	+2.2 pts of land/yr (p = 0.25)	Too noisy
5 Matching, 5 ways: look-alike areas picked by algorithm	All slower; fake-test p 0.15–0.60; no common support	Can't decide
6 Synthetic DiD: zones and years weighted to match Tambopata's past	−1,544; size rank 3 of 7	Holds up in ha

CAUSAL · CONDITIONAL

One causal range: 1,453–1,663 ha a year

Estimated change in net mining additions, ha/yr · Tambopata buffer zone · 2019–21 · causal only if the comparison zones show what Tambopata would have done; an upper bound; part of it is pits stopping

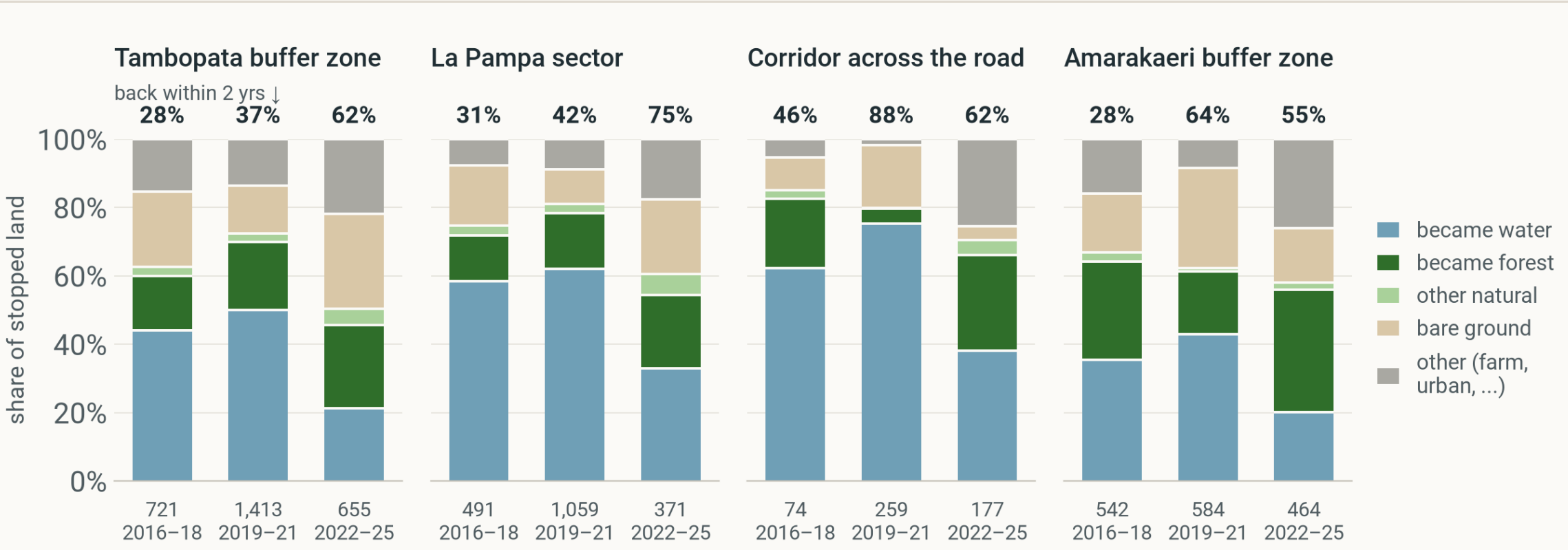


Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. 15 Peruvian buffer zones, 2014–25 (zone + year effects and synthetic DiD); Tambopata vs Amarakaeri on the 2014–18 baseline. Fake-2016 test: same model with a fake start in 2016.

DESCRIPTION

Stopped land: often water, often mined again within 2 years

Stopped land (left the mining class), share by what it became and share mining again within 2 years, period means · four zones · 2016–18, 2019–21, 2022–25 (small numbers under the bars: stopped ha/yr). Numbers on top: share mining again within 2 years.



Source: MapBiomas Peru Collection 4 (30 m, class 4.2 mining, legal and illegal), our pixel accounting. Back within 2 years = classifier flicker or re-entry; the last two years cannot be followed up. No image validation sample.

DESCRIPTION

Road and 1-km cell designs: too noisy to tell

Net mining additions · La Pampa longitudes · 2019–21 vs 2014–18 · finer-grained designs, described in one line each

EITHER SIDE OF THE ROAD

+2.2 pts

of land per year swung north of the road (p = 0.25; 0.18–0.86 across distances). Compares land just either side of the highway; miners moving across break it, so it is an upper bound.

AT THE RESERVE EDGE

−0.2 pts

no measurable change (p = 0.63). Same comparison, inside vs outside the reserve's border.

1-KM CELLS

≈ before/after

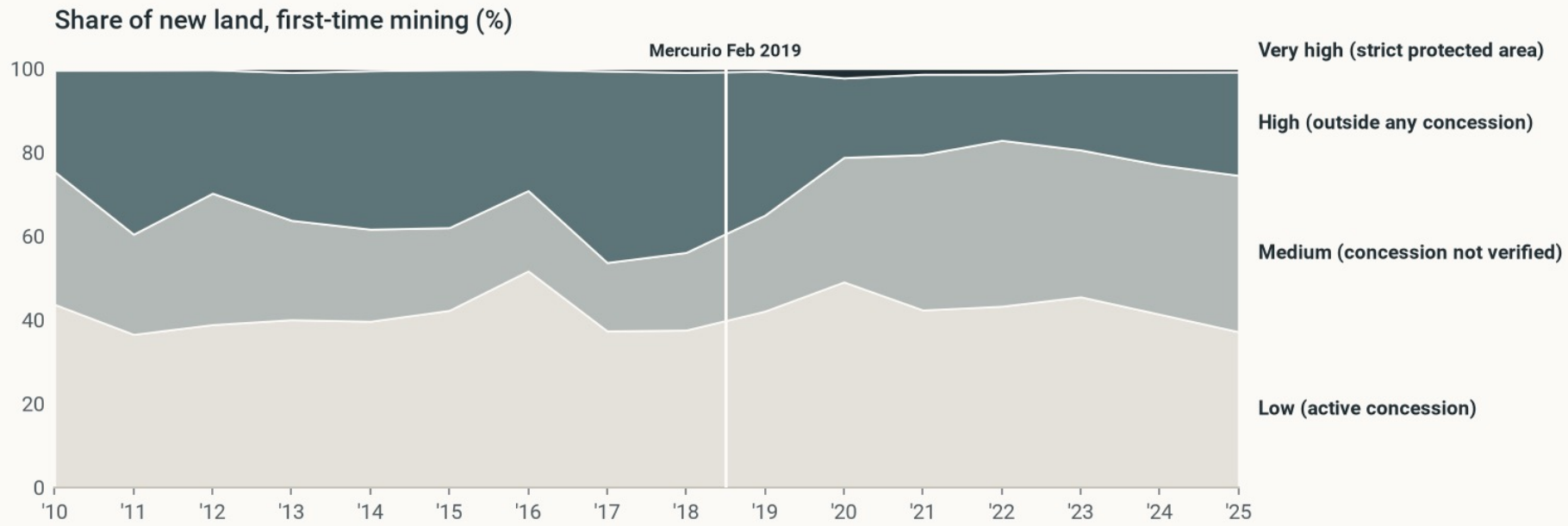
Each 1-km cell compared with cells of the same year; the comparison cells hold almost no mining, so this is close to a simple before/after.

DESCRIPTION

Presumed-illegal new land fell; grey zones grew

Presumed-illegal new land fell from 41% to 21% of first-time mining

New land (first-time mining) in Madre de Dios by MAAP legality level, share of each year's hectares (%), 2010–2025.
High + Very high: 41% → 25% → 21%



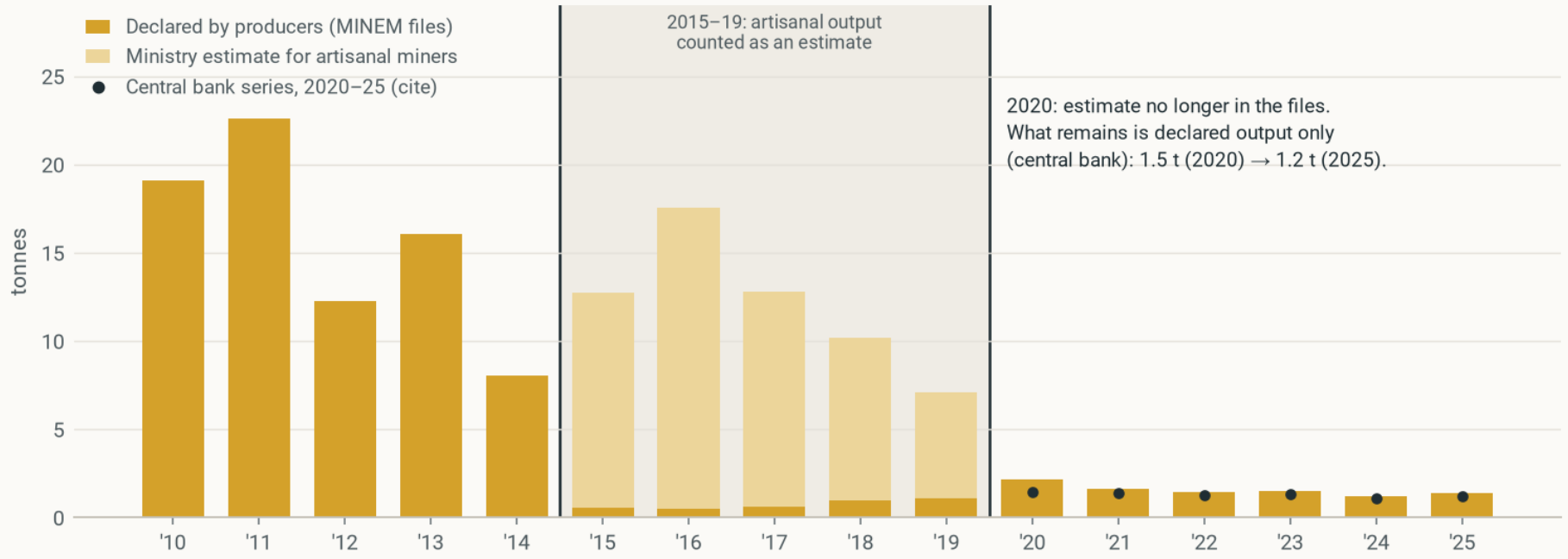
Source: Amazon Mining Watch / MAAP #235 legality layer (GitHub earthrise-media/mining-detector) x MapBiomas Peru Col 4 first-mined year (new land), zone registry. Today's concessions applied to all years (a time-consistent version using filing dates gives the same fall). Description.

DESCRIPTION

The official gold drop is a definition change

Official gold 'fell' in 2020 because the ministry stopped adding its artisanal estimate

Madre de Dios gold in the official series, tonnes per year: what producers declared plus MINEM's estimate



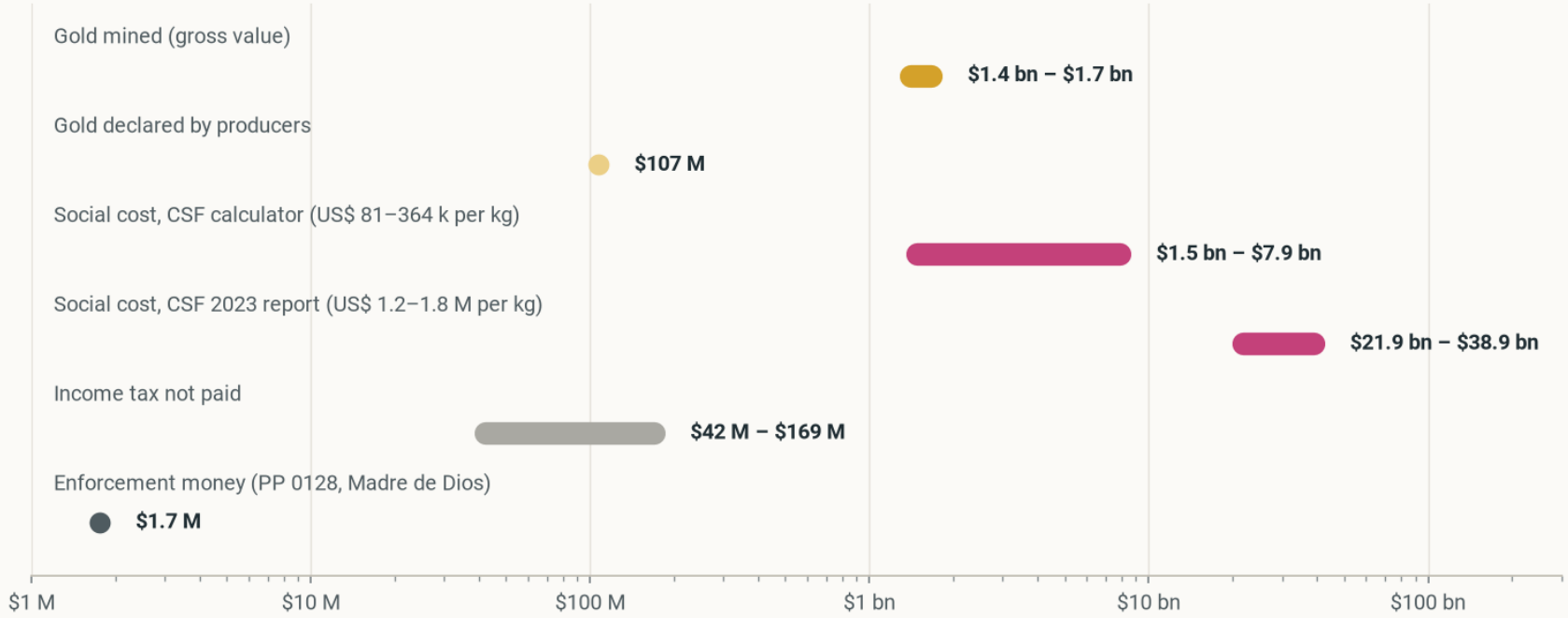
Source: MINEM, Producción minera (gob.pe 5472883), yearly gold files; Madre de Dios rows; BCRP series RD15556DA. Both parts sum to the central bank series in 2005-14 and 2017-19 (within 2%; every year listed in output/minem_vs_bcrp.csv), not in 2001-04, 2015-16 nor from 2020, when the MINEM files run above the central bank: for declared output after 2020 we cite the central bank (dots). Description.

PREDICTION

Gold and its damage dwarf enforcement money

Each year, mining's damage and gold dwarf the money spent stopping it

Madre de Dios, per year, 2022–25, US$ (log scale). Bars = low–high range from cited parameters. Informed speculation, not measurement

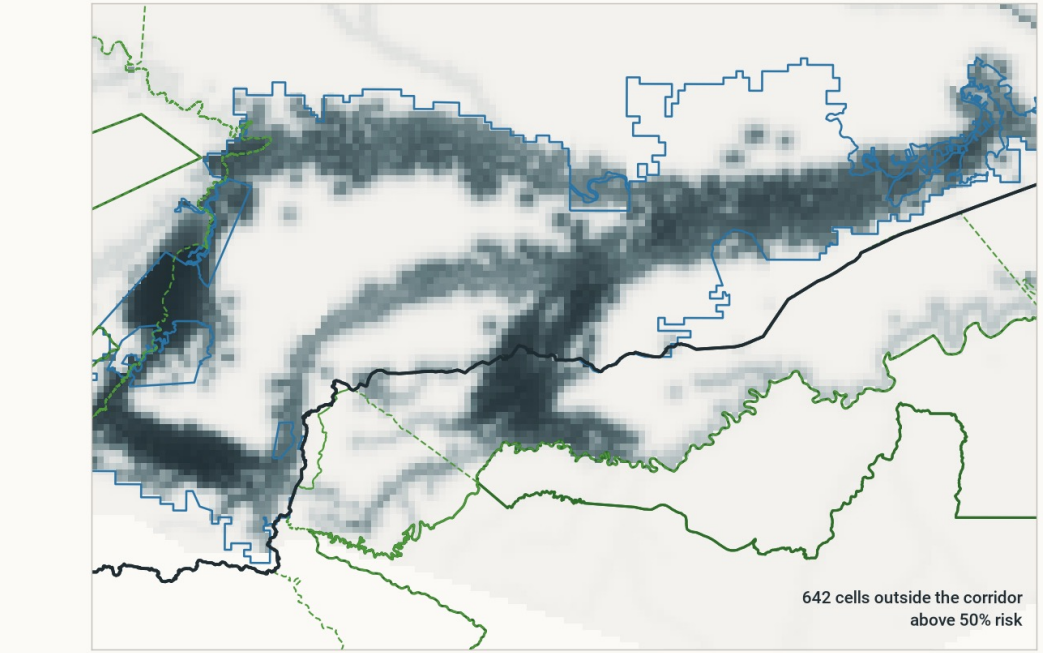


Sources: our new land, first-time mined hectares (MapBiomas Peru Col 4; old pits given no gold; not net additions) x CSF (2023) 1.6–1.9 kg gold/ha; World Bank gold price; MINEM declared gold; income tax 29.5% x 11–37% margin (D.L. 1269; Torres 2015); social cost: two CSF estimates shown separately; MEF PP 0128 at BCRP annual exchange rates. Full citations: docs/walkthrough/notes/economics_sources.md.

Next new mining is likely next to old mining

New land is predicted where mining already is (held-out AUC 0.92 vs 0.91 for 'mining nearby' alone)

Predicted chance of more than 1 ha of new land (first-time mining) per 1-km cell in 2026–28. Prediction.

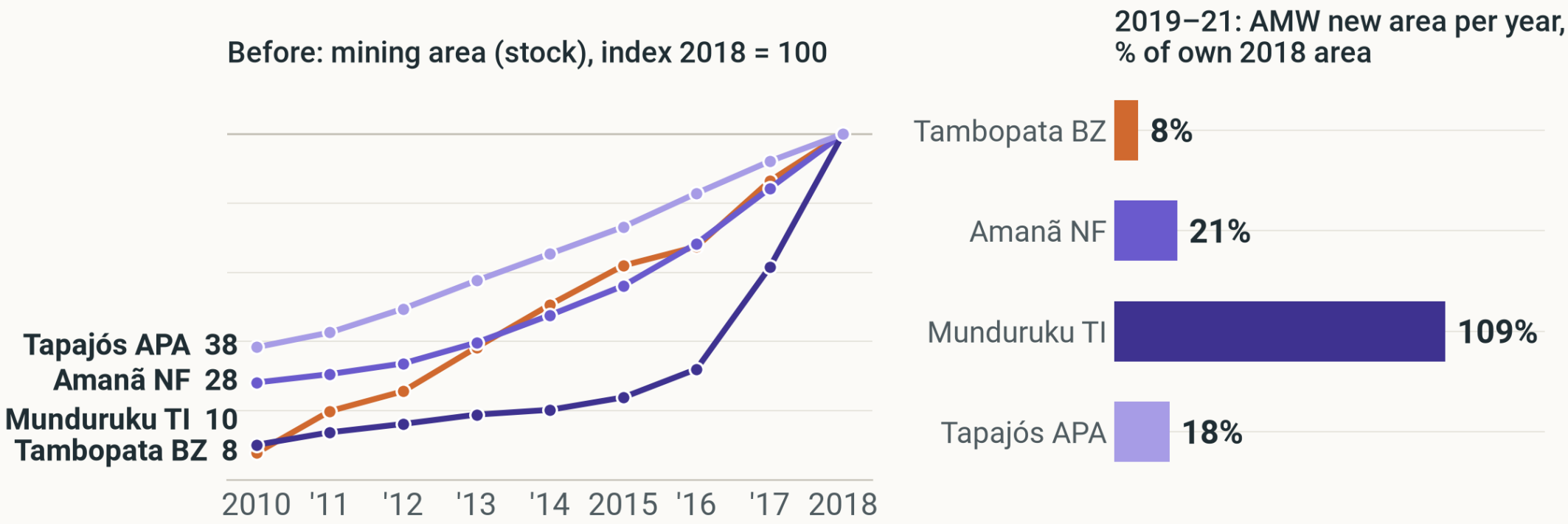


Source: MapBiomas Peru Collection 4 rasters (class 4.2 Minería), our pixel accounting, zone registry. Logit on 1-km cells: true distance to the highway, distance to rivers (2018), gold-bearing land, mining tested on 2022–25 in held-out 10-km blocks, applied to 2025 features. Reserves score low because they held almost no mining in training; river distance is in the model, but not river access by boat, so the 2025 Malinowski incursion inside the reserve is not foreseen. Prediction, not a causal map.

DESCRIPTION

Twins: Tambopata at 16%–41% of their pace; trends differed before

Left: mining area (stock), index 2018 = 100, 2010–18 (pre-trends rejected, p < 0.001). Right: Amazon Mining Watch new area 2019–21 per year, % of own 2018 area: a different measure, not net additions.

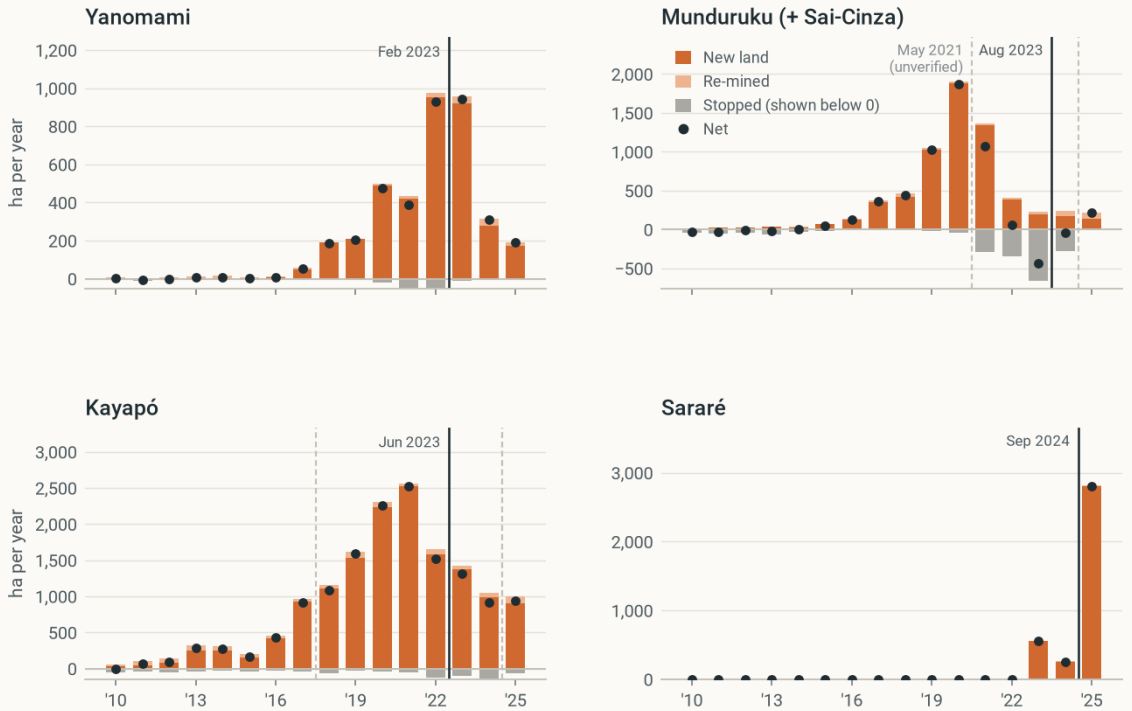


Source: MapBiomas Peru Col 4 rasters, v3 registry (Tambopata) / MapBiomas Amazonía Col 6 (twins); Amazon Mining Watch new area, area-weighted (comparable to first-time mining, not net). Twins chosen by fixed rules (Ruby's four rules). Different collections: compared as indices, never subtracted.

Brazil: mining boomed and fell; operations arrived at the peak

Brazil: mining boomed for years and fell after 2022; operations arrived at the peak

Net mining additions per year, ha (dot) = new land + re-mined (up) − stopped (down), MapBiomas Brazil Collection 11, 2010–2025.
Our data cannot separate the operations from the end of the boom.

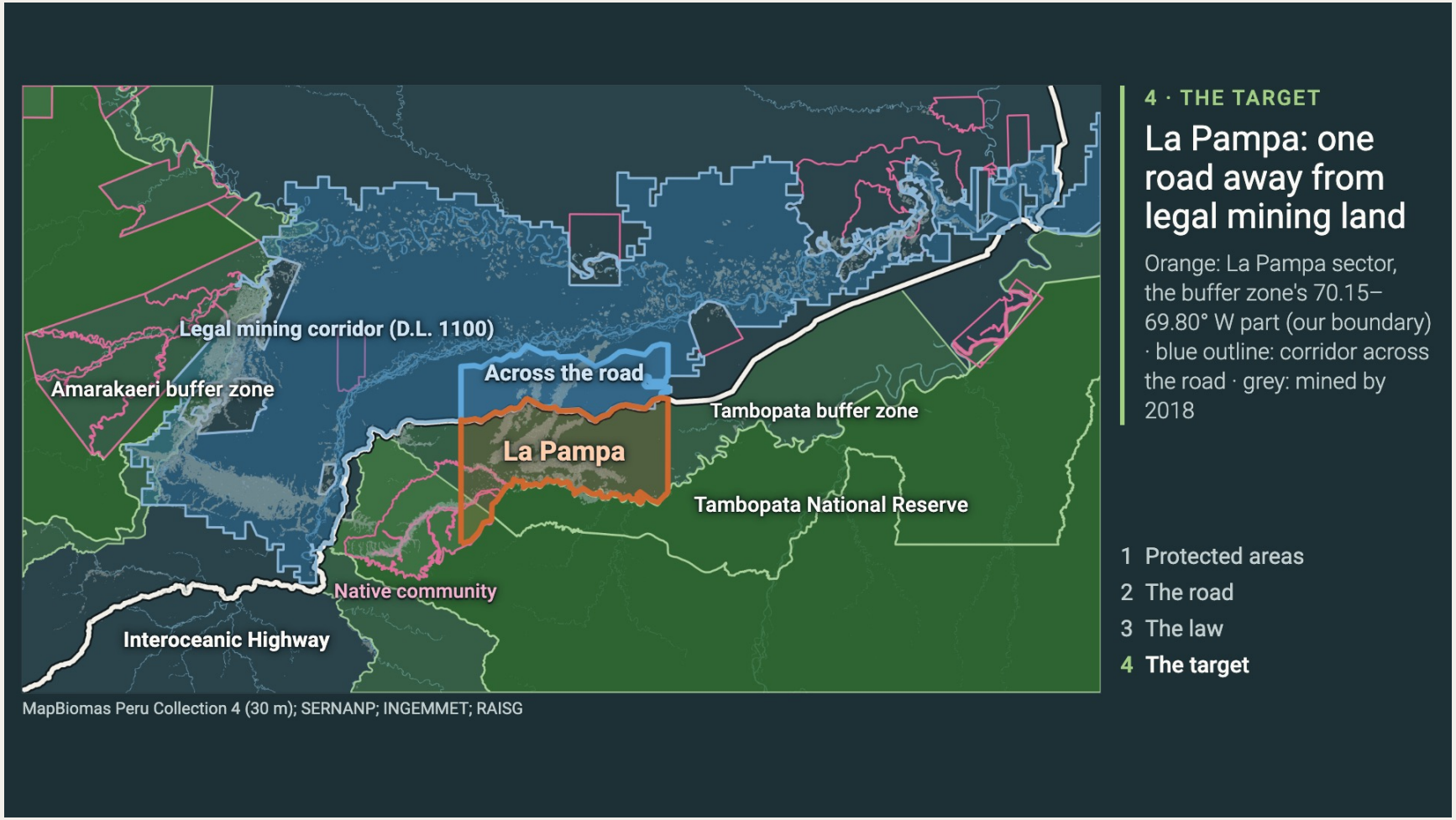


Source: MapBiomas Brazil Collection 11 rasters (class 30 mining) x FUNAI territories, our pixel accounting; operations: data_intermediate/operations_coded.csv (rule R*). Black line = the territory's first verified operation 2021–25 (rule R*), drawn before its first treated year (Jan–Jun: that year; Jul–Dec: the next). Dashed grey = other operations, not coded as treatment. Event-study leads (net, ha/yr): e = −3 +32, e = −2 +162, e ≤ −6 -396: not flat (the treated territories boomed before their operations; card 12 finds parallel trends fail). Y-axes differ. Description.

DESCRIPTION

The map build as one animation

Same four steps as the click-through map slides · protected areas → highway → corridor → La Pampa · click to play



DESCRIPTION

Definitions and data

One outcome, one set of zones, one treatment · every number in this deck comes from our replication code

ITEM	DEFINITION	SOURCE
Net mining additions	Mining area in year t − year t−1, ha/yr = new land + re-mined − stopped	MapBiomas Peru Collection 4 (30 m, class 4.2)
New land / re-mined / stopped	First time since 1985 / mined again after a gap / left the mining class	Our pixel accounting
Tambopata buffer zone	Official SERNANP polygon, whole (incl. Kotsimba)	SERNANP
La Pampa sector	Tambopata buffer zone between 70.15° and 69.80° W (no official line)	Our boundary
Corridor across the road	D.L. 1100 land ≤ 10 km north of the highway, at La Pampa's longitudes	INGEMMET, RAISG
Legal mining corridor	All D.L. 1100 land in Madre de Dios (native communities excluded); its share = its net additions ÷ the department's, summed over each period	INGEMMET
Treatment and periods	Mercurio from Feb 2019 (first treated year 2019) · 2016–18 before · 2019–21 Mercurio and what followed · 2022–25 later years (± 2023)	MAAP #130; course card